Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 16 GB of memory. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the f1 score of MobileNet by dataset.

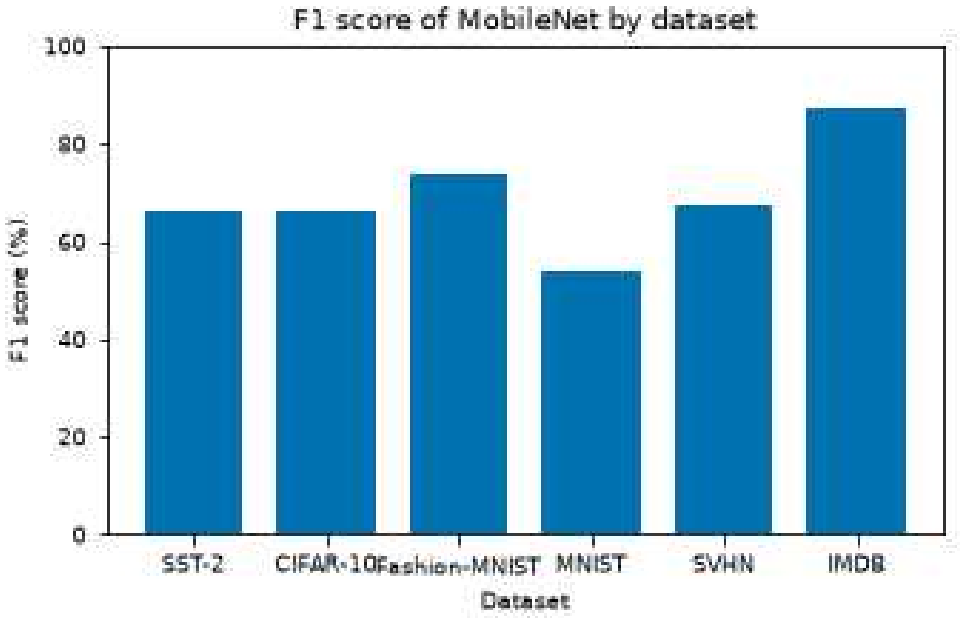


Figure 1: F1 score of MobileNet by dataset for the models in this experiment.

On SST-2, MobileNet obtains a F1 score of 61.4%. On MNIST, MobileNet obtains a F1 score of 54% (Figure 1).

Figure 2 shows the validation accuracy by dataset.

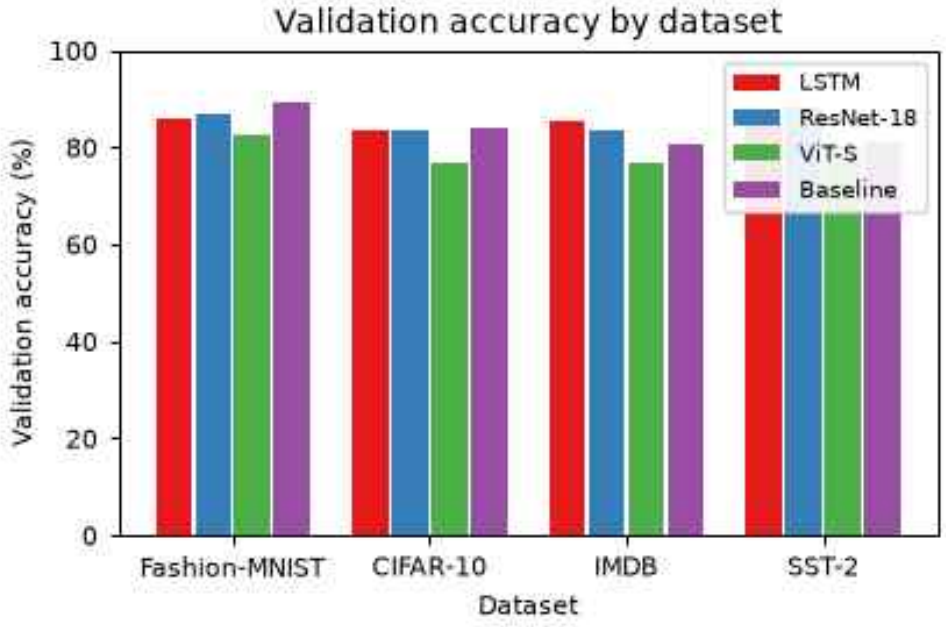


Figure 2: Validation accuracy by dataset for the models in this experiment.

Baseline outperforms ViT-S by 4 percentage points on IMDB (Figure 2). As shown in Figure 2, on SST-2,

ViT-S obtains a validation accuracy of 85.0%. As shown in Figure 2, on CIFAR-10, ViT-S obtains a validation accuracy of 83.5%. ViT-S reaches 80.5% validation accuracy on Fashion-MNIST.

Figure 3 shows the training loss.

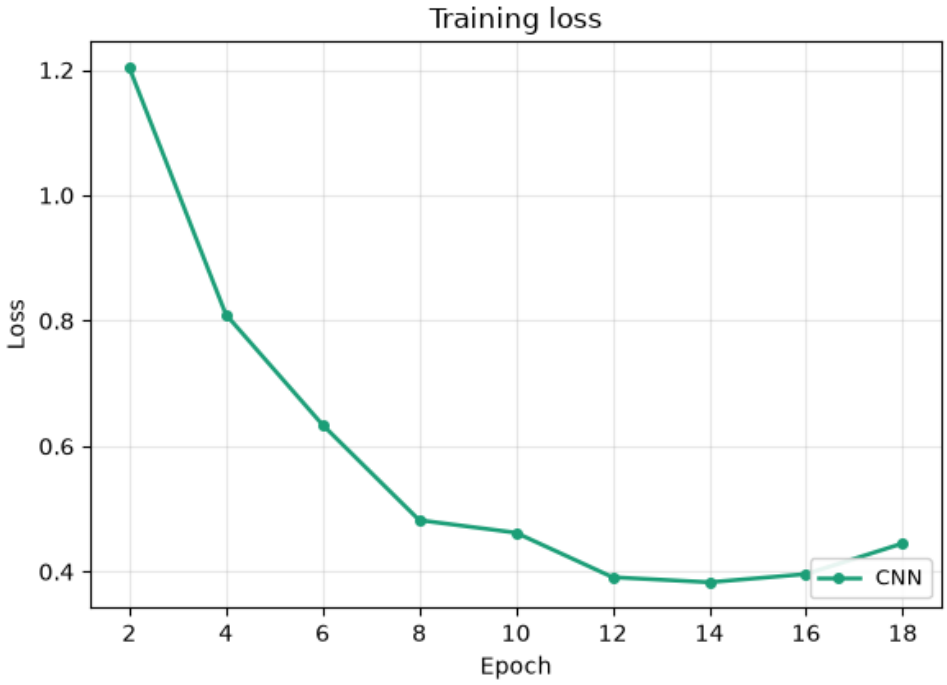


Figure 3: Training loss for the models in this experiment.

CNN has a loss of about 0.5 at epoch 10. CNN has a loss of 0.46 at epoch 10 (Figure 3). CNN has a loss of 0.634 at epoch 6.

Figure 4 shows the training loss.

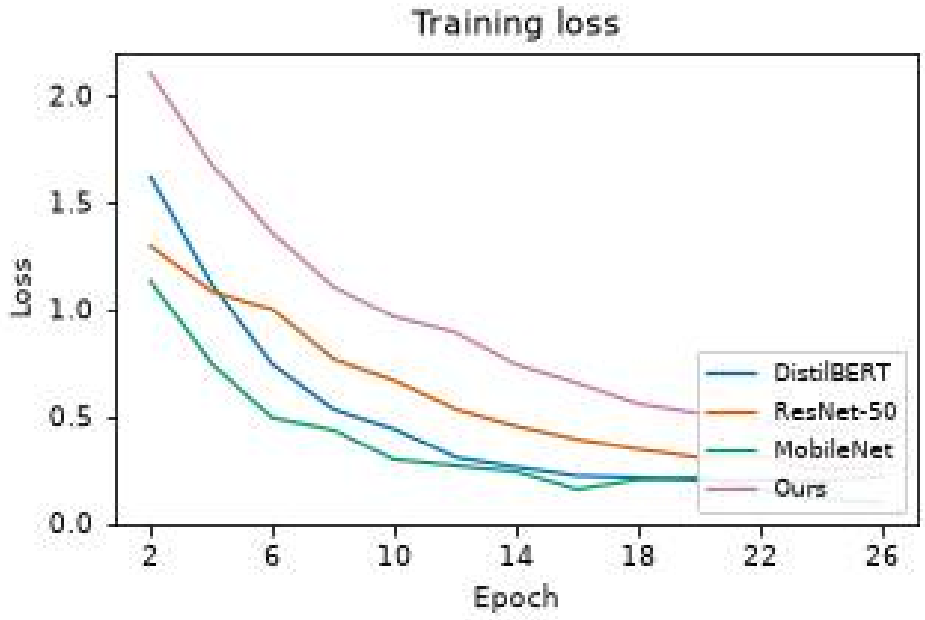


Figure 4: Training loss for the models in this experiment.

At epoch 16, the loss of DistilBERT is 0.218. Ours has a loss of 0.56 at epoch 18. As shown in Figure 4, ResNet-50 has a loss of over 0.2 at epoch 24. At epoch 4, the loss of Ours is 1.897.

3 Discussion

Logistic Regression reaches 80.1% accuracy on MNIST. We note that the training was stable for all models.